MIDDLE TECHNICAL UNIVERSITY

College of Engineering Technology

Department of Mechatronics Engineering Technology

ENGINEERING HOMEWORK REPORT

CONTROL THEORY

Lecture 2

Electrical Circuits

Mesh Analysis – Homework 1, 2 and 3

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Stage	Second Stage
Subject	Control Theory
Lecture	2
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Homework 1

Mesh Current Analysis

Question. Find the transfer function $G(s) = V_o(s)/V_i(s)$ of the circuit shown below.

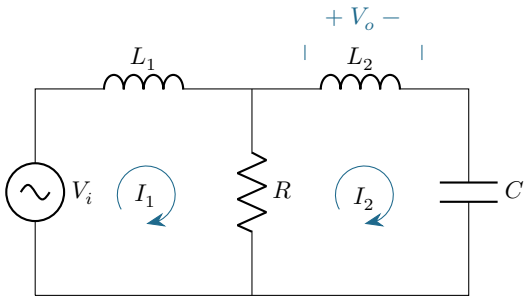


Figure 1: Two-mesh RLC circuit; V_o is measured across L_2.

1. Laplace-domain impedances

$$Z_{L_1} = L_1 s, \qquad Z_{L_2} = L_2 s, \qquad Z_R = R, \qquad Z_C = \frac{1}{Cs},$$
$$V(s) = Z(s)I(s), \qquad I_R = I_1 - I_2.$$

2. Mesh 1 (KVL)

$$\begin{aligned} V_i &= L_1 s I_1 + R(I_1 - I_2) \\ &= L_1 s I_1 + R I_1 - R I_2 \\ \boxed{(L_1 s + R) I_1 - R I_2} &= V_i. \end{aligned} \tag{1}$$

3. Mesh 2 (KVL)

$$\begin{aligned} 0 &= R(I_2 - I_1) + L_2 s I_2 + \frac{I_2}{Cs} \\ &= R I_2 - R I_1 + L_2 s I_2 + \frac{I_2}{Cs} \\ \boxed{-R I_1 + \left(R + L_2 s + \frac{1}{Cs}\right) I_2} &= 0. \end{aligned} \tag{2}$$

4. Eliminate I_1

$$
\begin{aligned}
-RI_1 + \left(R + L_2 s + \frac{1}{Cs}\right) I_2 = 0 \\
-RI_1 &= -\left(R + L_2 s + \frac{1}{Cs}\right) I_2 \\
RI_1 &= \left(R + L_2 s + \frac{1}{Cs}\right) I_2 \\
I_1 &= \frac{R + L_2 s + \frac{1}{Cs}}{R} I_2.
\end{aligned}
$$

$$
\begin{aligned}
V_i &= (L_1 s + R) I_1 - R I_2 \\
&= (L_1 s + R) \left(\frac{R + L_2 s + \frac{1}{Cs}}{R} I_2 \right) - R I_2 \\
&= \frac{(L_1 s + R)(R + L_2 s + \frac{1}{Cs})}{R} I_2 - R I_2 \\
&= \left[\frac{(L_1 s + R)(R + L_2 s + \frac{1}{Cs})}{R} - R \right] I_2.
\end{aligned}
$$

5. Current ratio and output voltage

$$
\begin{aligned}
R &= \frac{R^2}{R} \\
V_i &= \left[\frac{(L_1 s + R)(R + L_2 s + \frac{1}{Cs}) - R^2}{R} \right] I_2 \\
RV_i &= \left[(L_1 s + R)(R + L_2 s + \frac{1}{Cs}) - R^2 \right] I_2.
\end{aligned}
$$

$$
\begin{aligned}
I_2 &= \frac{RV_i}{(L_1 s + R)(R + L_2 s + \frac{1}{Cs}) - R^2} \\
\frac{I_2}{V_i} &= \frac{R}{(L_1 s + R)(R + L_2 s + \frac{1}{Cs}) - R^2}.
\end{aligned}
$$

$$
\begin{aligned}
V_o &= Z_{L_2} I_2 \\
&= L_2 s I_2 \\
\frac{V_o}{V_i} &= \frac{L_2 s I_2}{V_i} \\
&= L_2 s \frac{I_2}{V_i} \\
&= L_2 s \left[\frac{R}{(L_1 s + R)(R + L_2 s + \frac{1}{Cs}) - R^2} \right] \\
&= \frac{R L_2 s}{(L_1 s + R)(R + L_2 s + \frac{1}{Cs}) - R^2}.
\end{aligned}
$$

6. Expand and simplify

$$
\begin{aligned}
D(s) &= (L_1 s + R)\left(R + L_2 s + \frac{1}{Cs}\right) - R^2 \\
&= L_1 s\left(R + L_2 s + \frac{1}{Cs}\right) + R\left(R + L_2 s + \frac{1}{Cs}\right) - R^2 \\
&= L_1 R s + L_1 L_2 s^2 + \frac{L_1}{C} + R^2 + R L_2 s + \frac{R}{Cs} - R^2 \\
&= L_1 R s + R L_2 s + L_1 L_2 s^2 + \frac{L_1}{C} + \frac{R}{Cs} \\
&= L_1 L_2 s^2 + R(L_1 + L_2)s + \frac{L_1}{C} + \frac{R}{Cs}.
\end{aligned}
$$

$$
\begin{aligned}
\frac{V_o}{V_i} &= \frac{R L_2 s}{L_1 L_2 s^2 + R(L_1 + L_2)s + \frac{L_1}{C} + \frac{R}{Cs}} \\
&= \frac{R L_2 s \cdot Cs}{[L_1 L_2 s^2 + R(L_1 + L_2)s + \frac{L_1}{C} + \frac{R}{Cs}]Cs}.
\end{aligned}
$$

$$
\begin{aligned}
R L_2 s \cdot Cs &= R L_2 C s^2 \\
L_1 L_2 s^2 \cdot Cs &= L_1 L_2 C s^3 \\
R(L_1 + L_2)s \cdot Cs &= RC(L_1 + L_2)s^2 \\
\frac{L_1}{C} \cdot Cs &= L_1 s \\
\frac{R}{Cs} \cdot Cs &= R.
\end{aligned}
$$

$$
\begin{aligned}
\frac{V_o}{V_i} &= \frac{R L_2 C s^2}{L_1 L_2 C s^3 + RC(L_1 + L_2)s^2 + L_1 s + R} \\
&= \frac{L_2 C s^2}{\frac{L_1 L_2 C}{R} s^3 + (L_1 + L_2)Cs^2 + \frac{L_1}{R} s + 1}.
\end{aligned}
$$

FINAL TRANSFER FUNCTION

$$
\boxed{\frac{V_o(s)}{V_i(s)} = \frac{L_2 C s^2}{\frac{L_1 L_2 C}{R} s^3 + (L_1 + L_2)Cs^2 + \frac{L_1}{R} s + 1}}
$$

Homework 2

Mesh Current Analysis

Question. Find the transfer function $G(s) = E_o(s)/E_i(s)$ of the circuit shown below.

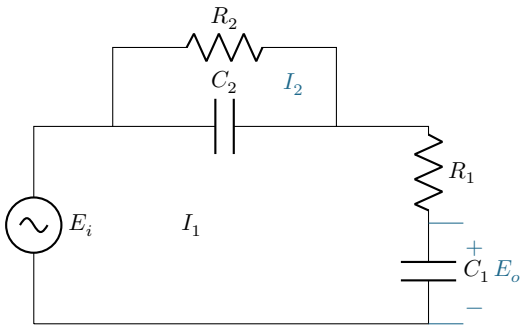


Figure 2: Two-mesh RC circuit; E_o is measured across C_1.

1. Laplace-domain impedances

$$Z_{R_1} = R_1, \qquad Z_{R_2} = R_2, \qquad Z_{C_1} = \frac{1}{C_1 s}, \qquad Z_{C_2} = \frac{1}{C_2 s},$$
$$V(s) = Z(s) I(s), \qquad I_{C_2} = I_1 - I_2.$$

2. Mesh 1 (KVL)

$$\begin{aligned} E_i &= \frac{I_1 - I_2}{C_2 s} + R_1 I_1 + \frac{I_1}{C_1 s} \\ &= \frac{I_1}{C_2 s} - \frac{I_2}{C_2 s} + R_1 I_1 + \frac{I_1}{C_1 s} \\ \boxed{\left(R_1 + \frac{1}{C_1 s} + \frac{1}{C_2 s}\right) I_1 - \frac{I_2}{C_2 s}} &= E_i. \end{aligned} \tag{1}$$

3. Mesh 2 (KVL)

$$\begin{aligned} 0 &= R_2 I_2 + \frac{I_2 - I_1}{C_2 s} \\ &= R_2 I_2 + \frac{I_2}{C_2 s} - \frac{I_1}{C_2 s} \\ \boxed{-\frac{I_1}{C_2 s} + \left(R_2 + \frac{1}{C_2 s}\right) I_2} &= 0. \end{aligned} \tag{2}$$

4. Eliminate I_2

$$
\begin{aligned}
-\frac{I_1}{C_2 s} + \left(R_2 + \frac{1}{C_2 s}\right) I_2 &= 0 \\
-I_1 + (R_2 C_2 s + 1)\, I_2 &= 0 \\
I_1 &= (R_2 C_2 s + 1)\, I_2 \\
\frac{I_1}{R_2 C_2 s + 1} &= I_2 \\
I_2 &= \frac{I_1}{R_2 C_2 s + 1}.
\end{aligned}
$$

5. Substitute into Mesh 1

$$
\begin{aligned}
E_i &= \left(R_1 + \frac{1}{C_1 s} + \frac{1}{C_2 s}\right) I_1 - \frac{I_2}{C_2 s} \\
&= \left(R_1 + \frac{1}{C_1 s} + \frac{1}{C_2 s}\right) I_1 - \frac{1}{C_2 s}\left(\frac{I_1}{R_2 C_2 s + 1}\right) \\
&= \left[R_1 + \frac{1}{C_1 s} + \frac{1}{C_2 s} - \frac{1}{C_2 s (R_2 C_2 s + 1)}\right] I_1.
\end{aligned}
$$

$$
\begin{aligned}
\frac{1}{C_2 s} - \frac{1}{C_2 s (R_2 C_2 s + 1)} &= \frac{R_2 C_2 s + 1}{C_2 s (R_2 C_2 s + 1)} - \frac{1}{C_2 s (R_2 C_2 s + 1)} \\
&= \frac{R_2 C_2 s + 1 - 1}{C_2 s (R_2 C_2 s + 1)} \\
&= \frac{R_2 C_2 s}{C_2 s (R_2 C_2 s + 1)} \\
&= \frac{R_2}{R_2 C_2 s + 1}.
\end{aligned}
$$

$$
\begin{aligned}
E_i &= \left[R_1 + \frac{1}{C_1 s} + \frac{R_2}{R_2 C_2 s + 1}\right] I_1 \\
\frac{E_i}{I_1} &= R_1 + \frac{1}{C_1 s} + \frac{R_2}{R_2 C_2 s + 1} \\
\frac{I_1}{E_i} &= \frac{1}{R_1 + \frac{1}{C_1 s} + \frac{R_2}{R_2 C_2 s + 1}}.
\end{aligned}
$$

6. Output voltage and transfer function

$$
\begin{aligned}
E_o &= Z_{C_1} I_1 \\
&= \frac{I_1}{C_1 s} \\
\frac{E_o}{E_i} &= \frac{I_1}{C_1 s E_i} \\
&= \frac{1}{C_1 s} \frac{I_1}{E_i} \\
&= \frac{1}{C_1 s} \left[\frac{1}{R_1 + \frac{1}{C_1 s} + \frac{R_2}{R_2 C_2 s + 1}} \right] \\
&= \frac{\frac{1}{C_1 s}}{R_1 + \frac{1}{C_1 s} + \frac{R_2}{R_2 C_2 s + 1}}.
\end{aligned}
$$

$$
\begin{aligned}
\frac{E_o}{E_i} &= \frac{1}{C_1 s \left(R_1 + \frac{1}{C_1 s} + \frac{R_2}{R_2 C_2 s + 1} \right)} \\
&= \frac{1}{R_1 C_1 s + 1 + \frac{R_2 C_1 s}{R_2 C_2 s + 1}} \\
&= \frac{R_2 C_2 s + 1}{(R_1 C_1 s + 1)(R_2 C_2 s + 1) + R_2 C_1 s}.
\end{aligned}
$$

7. Expand denominator

$$
\begin{aligned}
D(s) &= (R_1 C_1 s + 1)(R_2 C_2 s + 1) + R_2 C_1 s \\
&= R_1 C_1 s (R_2 C_2 s + 1) + 1(R_2 C_2 s + 1) + R_2 C_1 s \\
&= R_1 R_2 C_1 C_2 s^2 + R_1 C_1 s + R_2 C_2 s + 1 + R_2 C_1 s \\
&= R_1 R_2 C_1 C_2 s^2 + (R_1 C_1 + R_2 C_2 + R_2 C_1) s + 1.
\end{aligned}
$$

$$
\begin{aligned}
\frac{E_o}{E_i} &= \frac{R_2 C_2 s + 1}{D(s)} \\
&= \frac{R_2 C_2 s + 1}{R_1 R_2 C_1 C_2 s^2 + (R_1 C_1 + R_2 C_2 + R_2 C_1) s + 1}.
\end{aligned}
$$

FINAL TRANSFER FUNCTION

$$
\boxed{\frac{E_o(s)}{E_i(s)} = \frac{R_2 C_2 s + 1}{R_1 R_2 C_1 C_2 s^2 + (R_1 C_1 + R_2 C_2 + R_2 C_1) s + 1}}
$$

Homework 3

Mesh Current Analysis

Question. Find the transfer function $G(s) = I_2(s)/E_i(s)$ of the circuit shown below.

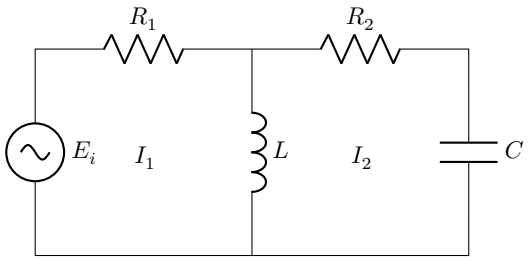


Figure 3: Two-mesh RLC circuit; the requested output is mesh current I_2.

1. Laplace-domain impedances

$$Z_{R_1} = R_1, \qquad Z_{R_2} = R_2, \qquad Z_L = Ls, \qquad Z_C = \frac{1}{Cs},$$
$$V(s) = Z(s)I(s), \qquad I_L = I_1 - I_2.$$

2. Mesh 1 (KVL)

$$\begin{aligned} E_i &= R_1 I_1 + Ls(I_1 - I_2) \\ &= R_1 I_1 + LsI_1 - LsI_2 \\ \boxed{(R_1 + Ls)I_1 - LsI_2} &= E_i. \end{aligned} \tag{1}$$

3. Mesh 2 (KVL)

$$\begin{aligned} 0 &= Ls(I_2 - I_1) + R_2 I_2 + \frac{I_2}{Cs} \\ &= LsI_2 - LsI_1 + R_2 I_2 + \frac{I_2}{Cs} \\ \boxed{-LsI_1 + \left(Ls + R_2 + \frac{1}{Cs}\right) I_2} &= 0. \end{aligned} \tag{2}$$

4. Eliminate I_1

$$\begin{aligned} -LsI_1 + \left(Ls + R_2 + \frac{1}{Cs}\right) I_2 &= 0 \\ -LsI_1 &= -\left(Ls + R_2 + \frac{1}{Cs}\right) I_2 \\ LsI_1 &= \left(Ls + R_2 + \frac{1}{Cs}\right) I_2 \\ I_1 &= \frac{Ls + R_2 + \frac{1}{Cs}}{Ls} I_2. \end{aligned}$$

5. Substitute into Mesh 1

$$
\begin{aligned}
E_i &= (R_1 + Ls)I_1 - LsI_2 \\
&= (R_1 + Ls)\left(\frac{Ls + R_2 + \frac{1}{Cs}}{Ls} I_2\right) - LsI_2 \\
&= \left[\frac{(R_1 + Ls)(Ls + R_2 + \frac{1}{Cs})}{Ls} - Ls\right] I_2.
\end{aligned}
$$

$$
\begin{aligned}
Ls &= \frac{L^2 s^2}{Ls} \\
E_i &= \left[\frac{(R_1 + Ls)(Ls + R_2 + \frac{1}{Cs})}{Ls} - \frac{L^2 s^2}{Ls}\right] I_2 \\
&= \left[\frac{(R_1 + Ls)(Ls + R_2 + \frac{1}{Cs}) - L^2 s^2}{Ls}\right] I_2.
\end{aligned}
$$

$$
\begin{aligned}
LsE_i &= \left[(R_1 + Ls)(Ls + R_2 + \frac{1}{Cs}) - L^2 s^2\right] I_2 \\
I_2 &= \frac{LsE_i}{(R_1 + Ls)(Ls + R_2 + \frac{1}{Cs}) - L^2 s^2} \\
\frac{I_2}{E_i} &= \frac{Ls}{(R_1 + Ls)(Ls + R_2 + \frac{1}{Cs}) - L^2 s^2}.
\end{aligned}
$$

6. Expand the denominator

$$
\begin{aligned}
D(s) &= (R_1 + Ls)\left(Ls + R_2 + \frac{1}{Cs}\right) - L^2 s^2 \\
&= R_1\left(Ls + R_2 + \frac{1}{Cs}\right) + Ls\left(Ls + R_2 + \frac{1}{Cs}\right) - L^2 s^2 \\
&= R_1 Ls + R_1 R_2 + \frac{R_1}{Cs} + L^2 s^2 + LR_2 s + \frac{L}{C} - L^2 s^2 \\
&= R_1 Ls + R_1 R_2 + \frac{R_1}{Cs} + LR_2 s + \frac{L}{C} \\
&= L(R_1 + R_2)s + R_1 R_2 + \frac{R_1}{Cs} + \frac{L}{C}.
\end{aligned}
$$

7. Clear fractions and collect terms

$$\frac{I_2}{E_i} = \frac{Ls}{L(R_1+R_2)s + R_1R_2 + \frac{R_1}{Cs} + \frac{L}{C}}$$
$$= \frac{Ls \cdot Cs}{[L(R_1+R_2)s + R_1R_2 + \frac{R_1}{Cs} + \frac{L}{C}]Cs}.$$

$$\begin{aligned} Ls \cdot Cs &= LCs^2 \\ L(R_1+R_2)s \cdot Cs &= LC(R_1+R_2)s^2 \\ R_1R_2 \cdot Cs &= R_1R_2Cs \\ \frac{R_1}{Cs} \cdot Cs &= R_1 \\ \frac{L}{C} \cdot Cs &= Ls. \end{aligned}$$

$$\frac{I_2}{E_i} = \frac{LCs^2}{LC(R_1+R_2)s^2 + R_1R_2Cs + R_1 + Ls}$$
$$= \frac{LCs^2}{(R_1+R_2)LCs^2 + (R_1R_2C + L)s + R_1}.$$

FINAL TRANSFER FUNCTION

$$\boxed{\frac{I_2(s)}{E_i(s)} = \frac{LCs^2}{(R_1+R_2)LCs^2 + (R_1R_2C + L)s + R_1}}$$